

Q5. Explain in your own words the difference between encapsulation and abstraction, using examples from any social media app you use daily (like Instagram or WhatsApp).

Ans: 1. Encapsulation – “Protecting the data”

Encapsulation means keeping data private and allowing access to it through controlled methods.

Instagram example:

Suppose an InstagramUser class has a private phoneNumber. Users should not be able to directly change the phone number from outside the class. Instead, we use methods like setPhoneNumber() and getPhoneNumber().

```
class InstagramUser {  
    private:  
        string phoneNumber;  
    public:  
        void setPhoneNumber(string number) {  
            phoneNumber = number;  
        }  
        string getPhoneNumber() {  
            return phoneNumber;  
        }  
};
```

So, encapsulation = protecting and controlling access to data.

2. Abstraction – “Hiding the complexity”

Abstraction means showing only the important features and hiding the complicated internal working.

Instagram example:

When you upload a photo, you simply click the Upload/Post button. You don't

need to know how Instagram compresses the image, stores it on servers, processes it, or delivers it to other users.

You just use something like:

```
uploadPhoto();
```

The complicated process happens internally.

So, abstraction = hiding unnecessary implementation details.